

Green's Functions in 2D via the Method of Images

Math Department

Free-Space Green's Function

For the Laplacian in $\mathbb{R}^2$

$$-\Delta G_0(x, \xi) = \delta(x - \xi)$$

the free-space Green's function is

$$G_0(x, \xi) = -\frac{1}{2\pi} \log |x - \xi|.$$

The **method of images** constructs Green's functions for domains by placing additional sources outside the domain so that boundary conditions are satisfied.

Half Space Geometry

Domain

$$\Omega = \{(x, y) : y > 0\}$$

Boundary

$$y = 0$$

Reflection of a point

$$\xi = (\xi_1, \xi_2)$$

across the boundary is

$$\xi^* = (\xi_1, -\xi_2).$$

A) Dirichlet Green's Function (Half Space)

For Dirichlet conditions

$$G = 0 \quad \text{on } y = 0$$

place a **negative image charge**.

$$G(x, \xi) = G_0(x, \xi) - G_0(x, \xi^*)$$

Thus

$$G(x, \xi) = -\frac{1}{2\pi} \log \frac{|x - \xi|}{|x - \xi^*|}.$$

On the boundary the two logarithms cancel.

B) Neumann Green's Function (Half Space)

For homogeneous Neumann conditions

$$\frac{\partial G}{\partial n} = 0$$

use a **positive image charge**.

$$G(x, \xi) = G_0(x, \xi) + G_0(x, \xi^*)$$

Reflection symmetry forces the normal derivative to vanish along the boundary.

C) Dirichlet on an Infinite Strip

Domain

$$0 < y < a$$

To enforce Dirichlet conditions on both boundaries we reflect the source repeatedly across both lines.

This produces an infinite array of alternating images.

$$G(x, \xi) = \sum_{k=-\infty}^{\infty} [G_0(x, \xi_k) - G_0(x, \xi_k^*)]$$

where

$$\xi_k = (\xi_1, \xi_2 + 2ka).$$

D) Neumann on an Infinite Strip

For Neumann conditions the reflections have the same sign.

$$G(x, \xi) = \sum_{k=-\infty}^{\infty} [G_0(x, \xi_k) + G_0(x, \xi_k^*)]$$

The periodic image lattice guarantees zero normal derivative on both boundaries.

Quadrant Geometry

Domain

$$x > 0, \quad y > 0$$

Two reflections:

$$(\xi_1, \xi_2)$$

$$(-\xi_1, \xi_2)$$

$$(\xi_1, -\xi_2)$$

$$(-\xi_1, -\xi_2)$$

E) Dirichlet in a Quadrant

Alternate the signs of the four sources:

$$G(x, \xi) = G_0(x, \xi) - G_0(x, \xi_1^*) - G_0(x, \xi_2^*) + G_0(x, \xi_{12}^*).$$

This ensures cancellation on both coordinate axes.

F) Neumann in a Quadrant

Use equal signs for the reflections:

$$G(x, \xi) = G_0(x, \xi) + G_0(x, \xi_1^*) + G_0(x, \xi_2^*) + G_0(x, \xi_{12}^*).$$

The symmetry forces vanishing normal derivatives on both boundaries.

Disk Geometry

Domain

$$|x| < R$$

Reflection is achieved by **Kelvin inversion**:

$$\xi^* = \frac{R^2}{|\xi|^2} \xi$$

The image point lies outside the circle.

G) Dirichlet in a Disk

The Green's function is

$$G(x, \xi) = -\frac{1}{2\pi} \left(\log |x - \xi| - \log \left| x - \frac{R^2}{|\xi|^2} \xi \right| \right)$$

plus a constant chosen so that

$$G = 0 \quad \text{on } |x| = R.$$

H) Neumann in a Disk

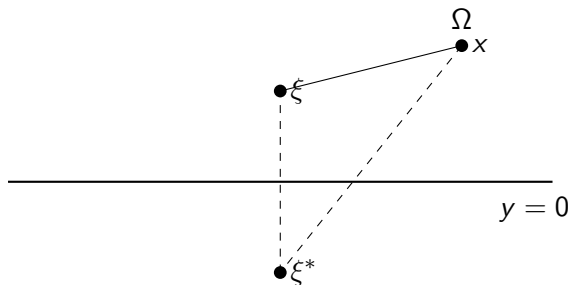
For Neumann conditions the Green's function has the form

$$G(x, \xi) = -\frac{1}{2\pi} \left(\log |x - \xi| + \log \left| x - \frac{R^2}{|\xi|^2} \xi \right| \right) + C$$

where the constant C ensures the compatibility condition

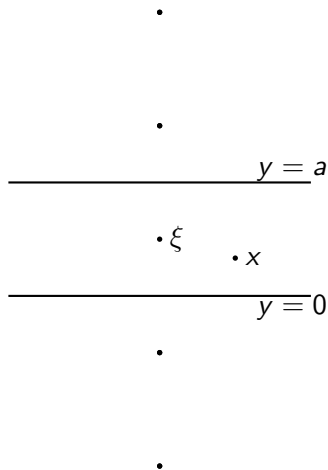
$$\int_{\partial\Omega} \frac{\partial G}{\partial n} dS = 0.$$

Image Construction: Half Space



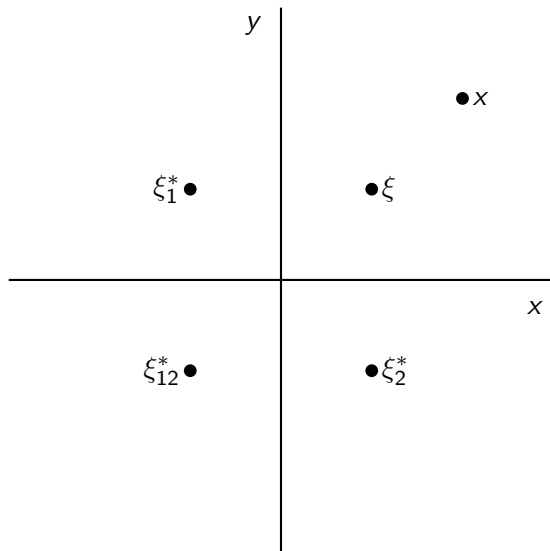
Reflection across the boundary produces the image source ξ^* .

Image Construction: Infinite Strip



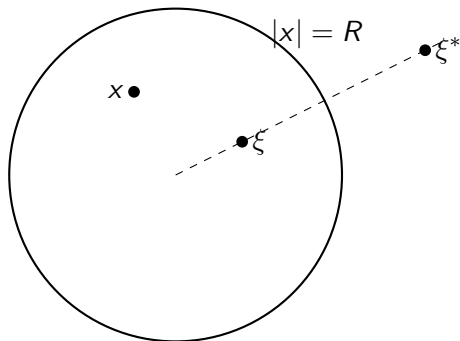
Repeated reflections create an infinite lattice of image sources.

Image Construction: Quadrant



Two reflections generate four image sources.

Image Construction: Disk



The image point lies on the same radial line and satisfies

$$\xi^* = \frac{R^2}{|\xi|^2} \xi.$$

Summary

Method of images constructs Green's functions by

- ▶ reflecting sources across boundaries
- ▶ choosing signs to enforce boundary conditions

Patterns:

- ▶ Dirichlet $\rightarrow$ alternating signs
- ▶ Neumann $\rightarrow$ equal signs
- ▶ Circular boundaries $\rightarrow$ inversion maps